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20 January 1976
SUPERSEDING
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MILITARY SPECIFICATION

SPECTACLE FRONTS AND TEMPLES, CELLULOSE ACETATE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers cellulose acetate plastic spectacle fronts with high positioned end pieces and butt joints at the temples, and cellulose acetate plastic spectacle temples (see 6.1).

1.2 Classification. Spectacle fronts and temples covered by this specification shall be of the following types and sizes, as specified (see 6.2).

Type I	Fronts, Male style, Military S-10 eyeshape (see referenced drawing)
Type II	Temples

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect, on date of invitation for bids or request for proposals, form a part of this specification, to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-397	Plastic Molding Material, Cellulose Acetate
L-P-504	Plastic Sheet and Film, Cellulose Acetate
QQ-C-585	Copper-Nickel-Zinc Alloy, Plate, Sheet, Strip, and Rolled Bar
QQ-C-586	Copper-Nickel-Zinc Alloy, Rod, Shapes, Flat Products with Finished Edges (Flat wire, Strip, and Bar)

FSC 6540

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PPP-B-566	Box, Folding, Paperboard
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiberboard Shipping
PPP-B-676	Box, Set-Up

MILITARY

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible

STANDARDS

FEDERAL

Fed. Test Method Std. No. 151	Metals; Test Methods
Fed. Test Method Std. No. 406	Plastics; Methods of Testing

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

DRAWING

Defense Personnel Support Center

24234	Spectacle, Front and Temple, Male Style, Cellulose Acetate

(Copies of specifications, standards, and drawing required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Standard	D 569
ASTM Standard	E 308

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

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AMERICAN NATIONAL STANDARDS INSTITUTE ANSI Z80.1 - 1972
AMERICAN NATIONAL STANDARD REQUIREMENTS FOR FIRST-QUALITY PRESCRIPTION OPHTHALMIC LENSES

(Application for copies should be addressed to American National Standards Institute, Inc., 1430 Broadway, New York, New York 10018).

3. REQUIREMENTS

3.1 Materials. Except as specified herein, materials shall conform to the applicable portions of the specifications, standards, and drawings specified. In addition, the metals shall conform to the chemical requirements of their applicable specifications when tested as specified in 4.4.2. When materials to be used are not specifically designated, those used shall be of the highest commercial quality and entirely suitable for the purpose intended.

3.1.1 Plastic parts.

3.1.1.1 Cellulose acetate plastic sheets. Cellulose acetate sheets shall be extruded and shall conform to type II of L-P-504 for the thickness specified in the drawing, and as specified herein, and shall be entirely suitable for the production of ophthalmic spectacle fronts and temples.

3.1.1.2 Cellulose acetate molding compound. The cellulose acetate material for injection molding ophthalmic spectacle fronts and temples shall be in conformity with L-P-397, Grade H2 or Grade H3. In addition, the material shall meet the additional requirements of L-P-504. The material shall be 100 percent virgin, with no admixture of scrap, sprues, runners, or other previously molded material.

3.1.1.3 Hardness. The hardness of the plastic fronts and temples shall be 95 minimum on the Rockwell R Hardness Scale. The test shall be conducted as described in 4.4.3.

3.1.1.4 Color. The color of the sheet stock from which the fronts and temples are blanked and the color of the test coupons produced by injection molding (3.1.1.2) shall be black.

3.1.1.4.1 Pigment. The pigment used for coloring shall be carbon black prepared by the impingement or furnace process. The percentage by weight of pigment shall be 0.09, with a tolerance of $\pm$ 2 percent. Coal tar derivatives or other dyestuff shall not be used (see 4.4.4).

3.1.1.4.2 Opacity. The finished plastic shall be opaque at a thickness of 0.140 inch when tested as specified in 4.4.5.

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3.1.1.5 Flow rate. The flow rate of the raw material (pellets) used to produce the sheet stock and for molding shall be H-2 or H-3, whichever will satisfactorily produce first quality spectacle components conforming to this specification (see 4.4.6).

3.1.1.6 Flammability. Plastic parts shall not burn faster than 24 seconds per inch when tested as specified in 4.4.7.

3.1.2 Metal parts. Metal for the plaque units shall be nickel silver conforming to composition 735 of QQ-C-585. Metal for the hinges and wire core shall be nickel silver conforming to composition 752, of QQ-C-585. Metal for the screws shall be nickel silver conforming to composition 794 of QQ-C-586. As an alternate, the metal for the screws shall be nickel silver having the following composition: Copper 60-63%; Nickel 11-13%; Lead .8 to 1.2% Manganese .50% maximum; Zinc, balance; (Traces, Iron, 1%; Tin, 1%, Aluminum .1%). The temper shall be suitable for the purpose intended.

3.2 Construction.

3.2.1 Type I - Fronts. Fronts shall be constructed as specified herein and as shown on the drawing. Material for the fronts shall be as specified in 3.1.1. Unless otherwise specified, the fronts shall conform to the requirements of the drawing for style, design, and dimensions of the specified size.

3.2.1.1 Style. The front shall be a male style spectacle front with the military S-10 eyeshape. The eyeshape shall conform to sheet 1 of the specified drawing. Variations of eyeshape shall be within the range permitted by the drawing. Variations of the horizontal and vertical aperture dimensions (B & C) within the tolerance given shall result in corresponding variation in the radii of the lens opening to maintain the required shape. The eyeshape shall comply with the test of 4.4.8, but in no case will an irregularly shaped eye aperture be acceptable.

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3.2.1.2 Front. The fronts shall be fabricated by master cam profiling, die cutting, or die stamping from cellulose acetate plastic sheet as specified in 3.1.1.1, or produced by the injection molding process from cellulose acetate molding compound as specified in 3.1.1.2. The color shall be as specified in 3.1.1.4. Eyewires, bridge and endpieces shall be integral parts of the front. Dimensions of the finished front shall be as shown on the drawing. Non-toleranced dimensions shall be within manufacturer's commercial standards for quality ophthalmic frame.

3.2.1.3 Nose pads. If the fronts are fabricated from sheet stock, the nose pads shall be made from cellulose acetate sheet as specified in 3.1.1.1, shall be of the color specified in 3.1.1.4, and shall be permanently cemented to the front. They shall be smoothly blended with the front. There shall be no chatter, file marks, or surface irregularities on the pads. The back of the nose pads (side facing the lens opening) may be made with a flat surface, in lieu of the concave surface shown on the drawing. Only one type of nose pad shall be on each front. If the fronts are produced by injection molding, the nose pads shall be molded integrally with the front. The back of the nose pad (the side facing the lens opening) may be provided with draft, to facilitate removal from the mold, in lieu of the concave curve shown in the drawing. However, the nose pads shall not extend into the lens opening so as to interfere with proper glazing of the front.

3.2.1.4 Hinges and plaque rivets. Each front shall be provided with two front hinge members, one right hand side and one left hand side, fabricated from nickel silver as specified in 3.1.2. The hinge members shall be of two-barrel construction as shown on the drawing, and shall properly mesh with the three-barrel temple hinge member of the spectacle temples to form a complete hinge. The hinge members shall be securely held in place by means of double-pin rivet plaques. The plaques shall be made of nickel silver as specified in 3.1.2 and shall be fabricated as shown in the drawing. The rivets shall be spun, capped or flatheaded against the hinge plates. The plaques shall be firmly set against or slightly imbedded in the adjoining surface. Trade names on the plaques or special plaques are not acceptable. Hinges and plaques shall be polished; shall have a bright finish; and shall be free of prominent marks, polishing compound, sharp edges, burrs, and dirt which detract from their appearance.

3.2.2 Type II - Temples. Temples shall be constructed as specified herein and as shown on the drawing. Material for the temples shall be as specified in 3.1.1. Unless otherwise specified, the temples shall conform to the requirements of the drawing for style, design, and dimensions for the specified size.

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3.2.2.1 Temple assembly. The temples shall be fabricated by master cam profiling, die cutting, or die stamping from cellulose acetate plastic sheet as specified in 3.1.1.1, or produced by the injection molding process from cellulose acetate molding compound as specified in 3.1.1.2. The color shall be as specified in 3.1.1.4.

3.2.2.1.1 Temples made from sheet stock shall be made either by the lamination of two sheets or equal thickness of sheet stock (3.1.1.1) with a nickel silver wire core as specified in 3.1.2, which shall be cemented or driven into the approximate center of the temple as shown in the drawing, or from solid (single thickness) sheet stock (3.1.1.1) with a nickel silver wire core as specified in 3.1.2, which shall be driven into the approximate center of the temple as shown in the drawing. As an alternate construction, the flat portion of the wire core may extend to the mitered end of the temple in lieu of the construction shown in the drawing. The flat end of the core shall be flush with the mitered end of the temple. Construction used shall comply with 3.2.2.2.

3.2.2.1.2 Temples made by the injection molding process from molding compound specified in 3.1.1.2 shall have a nickel silver wire core (3.1.2) which shall be driven or molded into the approximate center of the temple as shown in the drawing. As an alternate construction, the flat portion of the wire core may extend to the mitered end of the temple in lieu of the construction shown in the drawing. The flat end of the core shall be flush with the mitered end of the temple. Construction used shall comply with 3.2.2.2.

3.2.2.2 Core centering requirements. Upon visual inspection of the front edge, the core shall have a definite covering of plastic or when using alternate construction of 3.2.2.1.1 or 3.2.2.1.2 shall be visible and flush with the mitered end, the flat portion shall be parallel to the sides of the temple. If non-parallelism is observed, the angle of the edge of the core flat with the sides of the temple shall be a maximum of ten (10) degrees. No fractures, crazing, bulging, or other visible defects caused by the proximity of the core to the surfaces of the plastic shall be permitted. In the side view, the edges of the core, along its length, shall fall within two lines; each parallel to and 0.0625 inches from the centerline of the temple shown on the drawing.

3.2.2.3 Hinges and screws. Each right and left temple shall be provided with a corresponding right or left temple hinge member of three-barrel construction. They shall be fabricated from nickel silver as specified in 3.1.2 and as shown in the drawing. The hinge member shall mesh properly with a

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two-barrel front hinge member of the spectacle front as shown in the drawing to form a complete hinge. The temple hinge member shall have the bottom barrel tapped to accept the hinge screw. The top barrel shall be countersunk to accept the head of the screw. The temple hinge member shall be positioned as shown in the drawing and shall be securely fastened by means of double-pin, plaque rivets. The plaques shall be the same as specified in 3.2.1.4. The rivets shall be spun, capped, or flatheaded against the hinge plate. The rivets must go through the core without breaking the edge of the flat. The plaques shall be firmly set against or slightly embedded in the adjoining surface. Hinges and plaques shall be polished, shall have a bright finish, and shall be free of prominent marks, polishing compound, sharp edges, burrs, and dirt which detract from their appearance.

3.2.2.4 Temples shall be furnished in pairs, consisting of one right and one left temple of the size specified.

3.4 Identification marking.

3.4.1 Fronts. Each front shall be distinctly and neatly marked with the manufacturer's name or registered trademark, eye size, and bridge size. The markings shall be located as shown on the drawing.

3.4.2 Temples. Each temple shall be marked neatly, distinctly, and inconspicuously, with the manufacturer's name or registered trademark, the temple length, and the "joint-to-bend" measurement on the inside surface of the temple butt.

3.5 Finish. The fronts and temples shall be polished to a high gloss finish uniformly mirror-like, by means of manual or mechanical polishing. Chemical or dip methods of polishing shall not be acceptable. The gloss of the components shall be equal or superior to that of standard samples available from the procuring activity. Fronts and temples representative of lots submitted and accepted as First Article or pre-award samples may be used as standards.

3.6 Completed fronts. Completed fronts for testing purposes shall include metal parts.

3.6.1 Eyewire strength. The eyewires of the completed front shall comply with the requirements of 3.6.1.1 and 3.6.1.2.

3.6.1.1 Eyewire tensile strength. The load to break of each eyewire, regardless of designated eye size, shall be a minimum of 275 pounds. The tests shall be conducted as specified in 4.4.9 and 4.4.9.1.

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3.6.1.2 Eyewire compression test. Neither eyewire shall fracture or break when tested as described in 4.4.9 and 4.4.9.2.

3.6.2 Glazing, and adjustability. Each front sample shall be heated and glazed at least three times under normal conditions existing at military ophthalmic laboratories used in volume production of prescription spectacles, without detrimental effects, as follows:

(a) The fronts shall be capable of being properly reglazed with no loss of lens retention ability.

(b) The fronts shall be capable of being adjusted and of retaining the adjustments.

(c) There shall be no cracks, breaks or deformation in the front during and after lens insertion.

Tests shall be conducted as specified in 4.4.10. Lens retention and adjustment shall be evaluated after the samples have cooled to room temperature.

3.6.3 Gloss retention. There shall be no significant loss in the gloss after heating as specified in 4.4.11. In case of dispute the gloss of the front after heating shall equal or surpass that of the standard when tested as specified in 4.4.11.

3.7 Completed temples. Completed temples for testing purposes shall include the metal parts.

3.7.1 Adjustability. The completed temple samples shall be heated and adjusted at least three separate times under normal conditions at military opthalmic laboratories used in volume production and examined as follows:

(a) The temples shall be capable of being adjusted and of retaining the adjustments. Adjustment retention shall be evaluated after the adjusted temple has cooled to room temperature.

(b) There shall be no deleterious effects such as cracking or deformation.

Tests shall be conducted as specified in 4.4.10.

3.7.2 Gloss retention. There shall be no significant loss in the gloss after heating as specified in 4.4.11. In case of dispute the gloss of the temples shall equal or surpass that of the standard when tested as specified in 4.4.11.

3.8 Shape retention test. The front shall not exhibit permanent deformation exceeding 1 mm when tested as described in 4.4.12.

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3.9 Interchangeability. Fronts and temples furnished under this specification shall be interchangeable (see 6.1).

3.10 Workmanship. Workmanship shall be first class throughout. The finish on all spectacle components, shall be of first quality. The finish shall be a high gloss all over when viewed either individually or in small groups. The surfaces of the cellulose acetate shall be free from unsightly marks. There shall be no cracks, scratches, sharp edges, visible strains or weld lines, parting lines, knockout spots, sink marks, or other imperfections which detract from the appearance of the product or may impair strength or serviceability before and after glazing. The fronts and temples shall be clean and free of any polishing residue and dirt.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor. The certificate shall identify the material by lot number, date and the suppliers blend or formula number.

4.2 Quality conformance inspection. The quality conformance inspection shall consist of the examination and all the tests of this specification.

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4.2.1.1 Sample specimens (coupons) for determining physical properties of the plastic material. Sufficient samples (coupons) of the sheet stock material (3.1.1.1) or coupons made of the molding compound produced as specified in 3.1.1.2 shall be taken from each lot of material used in the manufacture of the items to determine compliance with all requirements contained in this specification. The unit of product for testing shall be the quantity necessary to perform all required tests one time each, taking into consideration the fact that, where possible, the same material may be used to perform two or more tests. In addition, the contractor shall keep on file certificates of quality or results of required tests performed by the contractor, covering the physical properties of each batch or lot of material utilized in the performance of the contract.

4.2.1.2 For chemical analysis of the metals. Four ounces of each lot of each metal used in the construction of the items shall be selected for chemical analysis as specified in 4.4.2.

4.2.1.3 End item sampling.

4.2.1.3.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The unit for examination shall be the unit of issue.

	Inspection level	AQL (percent defective)
For visual examination		
Major A	II	1.0
Major B	S-3	1.0
Minor A	II	4.0
Minor B	S-3	4.0
Minor C	S-3	6.5
For dimensional examination of hinges	2-from each lot of hinges	0

4.2.1.3.2 For tests. Sampling for each test shall be as indicated for inspection level S-1 in MIL-STD-105, except that the acceptance number shall be zero for all sample sizes, and a minimum of five samples shall be taken, Unless otherwise specified, the lot size for sampling for tests 4.4.9 Eyewire strength, 4.4.10, Glazing and adjustability, 4.4.11 Glass retention test and 4.4.12 Shape retention test shall be the production lot.

4.3 Examination. The fronts and temples shall be examined to determine compliance with all requirements contained in this specification.

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4.3.1 Classification of defects. Examination shall be conducted in accordance with, but not restricted to, the following classification of defects. Additional defects shall be classified in accordance with MIL-STD-105.

Categories and defects	
Major A	
101	Hinges not tightly fastened as determined by visual examination and finger manipulation.
102	Hinges damaged.
103	Cellulose acetate material not free of cracks.
104	Rivets not wholly located within flat of wire core.
105	Color not black.
Major B	
150	Eyewire groove not within tolerance.
151	Dimension "A" on the fronts not within tolerance.
152	Dimension "B" on the fronts not within tolerance.
153	Dimension "C" on the fronts not within tolerance.
154	When assembled to approved temples or fronts, as applicable, the downward angle ($10^{o} \pm 3^{o}$) of the temples is not within tolerance.
155	When assembled to the approved temples or fronts, as applicable, the setback angle ($96^{o} \pm 3^{o}$) of the temples is greater than specified.
156	$2\text{-}3/4^{o}$ Angle on butt end of temple greater than $4\text{-}3/4^{o}$.
157	$10^{o} \pm 2^{o}$ Angle butt end of temple not within tolerance.
158	Eyewire groove step less than 0.004 inch on back side.
Minor A	
201	Metal parts not free of polishing compound, dirt, prominent marks, sharp edges, and burrs.
202	Rivets not set with a smooth cap free of burrs.
203	Cellulose acetate material not free of polishing residue and dirt.
204	Fronts or temples not marked, or not properly marked, or marking not legible.
205	Groove in eyewire not centered.
206	Chatter marks in groove of eyewire.
207	Groove of eyewire irregular.

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Categories and defects (cont'd)

Minor A (cont'd)	
208	Groove of eyewire chipped at edges.
209	Cellulose acetate material not polished all over to a high gloss.
210	Metal hinges or plaques not polished to a bright finish.
211	Cellulose acetate material not free of scratches, sharp edges, visible strains, weld lines, flash, parting lines, knockout spots, sink marks, surface irregularities, and other imperfections.
212	Plaques not firmly set against or slightly imbedded in the adjoining surface.
213	Temples not paired in unit package.
214	Screw slot distorted.
215	Wire core not located as specified.
216	Screw missing.
217	Recess in end of screw missing.
Minor B	
250	Hinge plate not imbedded in the cellulose acetate material within tolerances specified.
251	When assembled to approved temples or fronts, as applicable, the setback angle ($96^{\circ} \pm 3^{\circ}$) of the temples is less than specified.
252	The 2-3/4° angle on the butt end of temple is less than 3/4°.
253	Size (fitting) dimension is incorrect.
254	Overall straight length of temple (Dimension "A") not within tolerances.
255	Screw threads damaged.
256	Core length not as specified.
Minor C	
301	Thickness of cellulose acetate material not within tolerance.
302	Face bow ($2^{\circ} \pm 2^{\circ}$) of front not within tolerance.
303	Mastoid bend (tip end) of paddle not within tolerance.
304	Down bend (ear bend) of temple paddle not within tolerance.
305	Width of temple at hinge not within tolerance.
306	Width of paddle tip not within tolerance.

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4.3.2 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing and marking requirements of section 5. Defects shall be scored as specified in table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with AQL of 4.0 expressed in terms of percent defective.

TABLE IV. Examination of preparation for delivery

Examine	Defect
Contents	not as specified.
Container	not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Material	Component missing or damaged.
Workmanship	Bulging or distoration of containers.

4.3.2.1 Examination of closure, waterproofing and banding of container. When the shipping container is required to comply with PPP-B-636, the examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the box specification.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

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4.4.1 Test conditions. Samples and specimens for test shall not be heated, cooled, immersed in water, or subjected to any mechanical or chemical treatment prior to test except as specifically described herein. Conditioning shall be at 23°± 1°C (73.5° ± 2°F) and 50 ± 5 percent Relative Humidity immediately prior to test.

4.4.2 Chemical analysis of the metals. The samples selected in accordance with 4.2.1.2 shall be analyzed in conformance with methods 111 or 112 of Federal Test Method Standard No. 151. In case of dispute, method 111 shall be used for referee analysis.

4.4.3 Hardness. Hardness of the finished fronts and temples shall be tested as specified in Federal Test Method Standard No. 406, method 1081, except specimen dimensions shall not apply. Hardness of test specimens (4.2.1.1) shall be acceptable. The manufacturer of the molding or extruding material certification is acceptable.

4.4.4 Carbon black. In lieu of certification by the molding material manufacturer, the percentage of carbon black shall be determined as follows:

Dissolve 10.00 to 12.00 grams of the molding material in a suitable solvent, such as acetone or methyl acetate. Steam bath heat or agitation over a 48 hour period shall be used to achieve complete solution. The solution shall be filtered through a tared fitted glass or Coors crucible (fine), dried and weighed. The carbon black on the crucible shall be calculated as percentage of the sample.

4.4.5 Opacity. The opacity of a 0.140 inch thick sheet or molded specimen shall be determined by use of a spectrophotometer or photometer. Transmitted light shall be zero. ASTM Standard E 308 shall be used for guidance.

4.4.6 Flow rate. Flow rate of the plastic shall be determined as described in ASTM Method D 569.

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4.4.7 Flammability. The fronts and temples shall be tested for determination of the flame propagation rate. Eyewires and temple samples shall be marked with a scribed line 1 inch from the end. The sample shall be ignited by holding the end of the 1 inch portion at the tip of a luminous 3/4 inch Bunsen burner flame. The time of burning to the 1 inch mark shall be determined by stopwatch and shall not be less than requirement of 3.1.1.6.

4.4.8 Eyeshape. The eyeshape of the front shall be checked with standard eye plug gauges as follows: Two flat pattern-type plug gauges shall be used. One shall be a "go" gauge made to the required shape and minimum dimensions of the particular eye size being examined. The other eye plug designated "no go" shall be the required shape and to the maximum dimension of the required eye size. The "go" plug gauge shall insert with ease or pass through the front and represent the shape of the aperture. The "no go" plug shall only be capable of insertion if slight pressure is used. The front and lot shall be rejected if the "go" gauge cannot be inserted or reveals shape differences. A few (two or three) minor gaps, up to 0.010 inch for spans up to 0.250 inch shall be acceptable if the "go" gauge is held firmly in place by the surrounding eyewire. The front and lot shall be rejected if the "no go" plug gauge inserts with ease and passes through the eye aperture.

4.4.8.1 Tooling. When required, layout and tooling shall be made available for government inspection.

4.4.9 Eyewire strength. The following tests shall be conducted using an Instron or similar laboratory testing machine.

4.4.9.1 Eyewire tensile test. Both eyewires shall be tested, but only one at a time. The eyewire shall be secured in the machine as shown in Figure 1 and subjected to a tensile load. The load application rate (speed of testing) shall be 1 inch per minute until the point of fracture. The load value shall comply with 3.6.1.1.

4.4.9.2 Eyewire compression test. The front, including both eyewires, shall be compressed so that the overall distance between the top and the bottom of the eyewires is reduced by 25 percent. The force shall be applied at a rate of 1 inch per minute. The fronts shall meet the requirements specified in 3.6.1.2.

4.4.10 Glazing and adjustability. Under laboratory conditions specified in 3.6.2 and 3.7.1, each front shall be heated, glazed and properly aligned 3 times by qualified personnel. The temples shall be heated and adjusted 3 times by qualified personnel. For acceptance testing, the procedure described in the first four sentences of 4.4.11 shall be used for preparing (heating) the samples used for glazing and adjustability tests. The test lens powers shall range from plano to a minimum of a minus three diopter sphere or sphero-cylinder. Lenses shall be glass. The lenses shall be edged to conform to the lens shape shown on the drawing for the front size being tested. Test lenses shall be correctly centered and aligned. Size tolerance shall be within $\pm$ 0.5 mm of that specified or as required by standard ANSI Z80.1. The bevel shall conform to that obtained from laboratory edgers, usually 110° to 125°.. When required, lenses or patterns may be obtained from the procuring activity.

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4.4.11 Gloss retention test. An American Optical Corporation Zyl Frame Warmer, or equivalent, shall be used for this test. The warmer shall be filled with glass beads (NSN 6640-00 299-9818), approximately 0.021 inch in diameter, to a depth of 1-1/2 to 1-3/4 inches. Temperature of the bead bath at the sample immersion depth shall be stabilized at 260° ± 10°F. The front or temple shall be completely submerged horizontally to a depth of 3/4 ± 1/4 inch for a period of 30 ± 2 seconds. The test sample shall be removed from the bath and allowed to cool for 5 minutes. The test samples shall be examined for gloss retention.

4.4.12 Shape retention test. Each front tested shall have the lens removed. The eyewire of the spectacle shall be heated to a temperature of 65° ± 3°C. and then stretched 2 mm. from its normal shape. It shall be held in this stretched position for a period of 30 minutes at ambient temperature. The eyewire is then released and allowed to recover for 24 hours. Permanent deformation shall not exceed 1 mm.

5. PACKAGING

5.1 Packaging requirements. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One pair of temples or each front, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Each front or each pair of temples shall be packaged in an envelope of suitable material and design. The dimensions of the envelope for each front shall not exceed 6-1/16 by 2-3/16 inches. The dimensions of the envelope for each pair of temples shall not exceed 6-1/4 by 2-5/16 inches. The flap of the envelope (closure) shall not be glued or sealed.

5.1.2.2 Intermediate package. Twelve unit packages containing fronts or temples of one size only shall be packed in a snug fitting box constructed in accordance with PPP-B-566 or PPP-B-676. Closure shall be adequate to prevent accidental opening.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

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5.2.1 Level A. Ninety six intermediate packages (1152 units) containing fronts or temples of one size only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard boxes shall be waterproofed as specified in the appendix of PPP-B-636.

5.2.2 Level B. Ninety-six intermediate packages (1152 units) containing fronts or temples of one size only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, constructed so as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2.4 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.3 Marking.

5.3.1 Interior packages. Each envelope and intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The fronts and temples are components of the standard military spectacle frame. When assembled, the frame is used to hold prescription lenses for corrective vision.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.3 Catalog information. The spectacle fronts and temples covered by this specification are listed in the Federal Supply Catalog under the following:

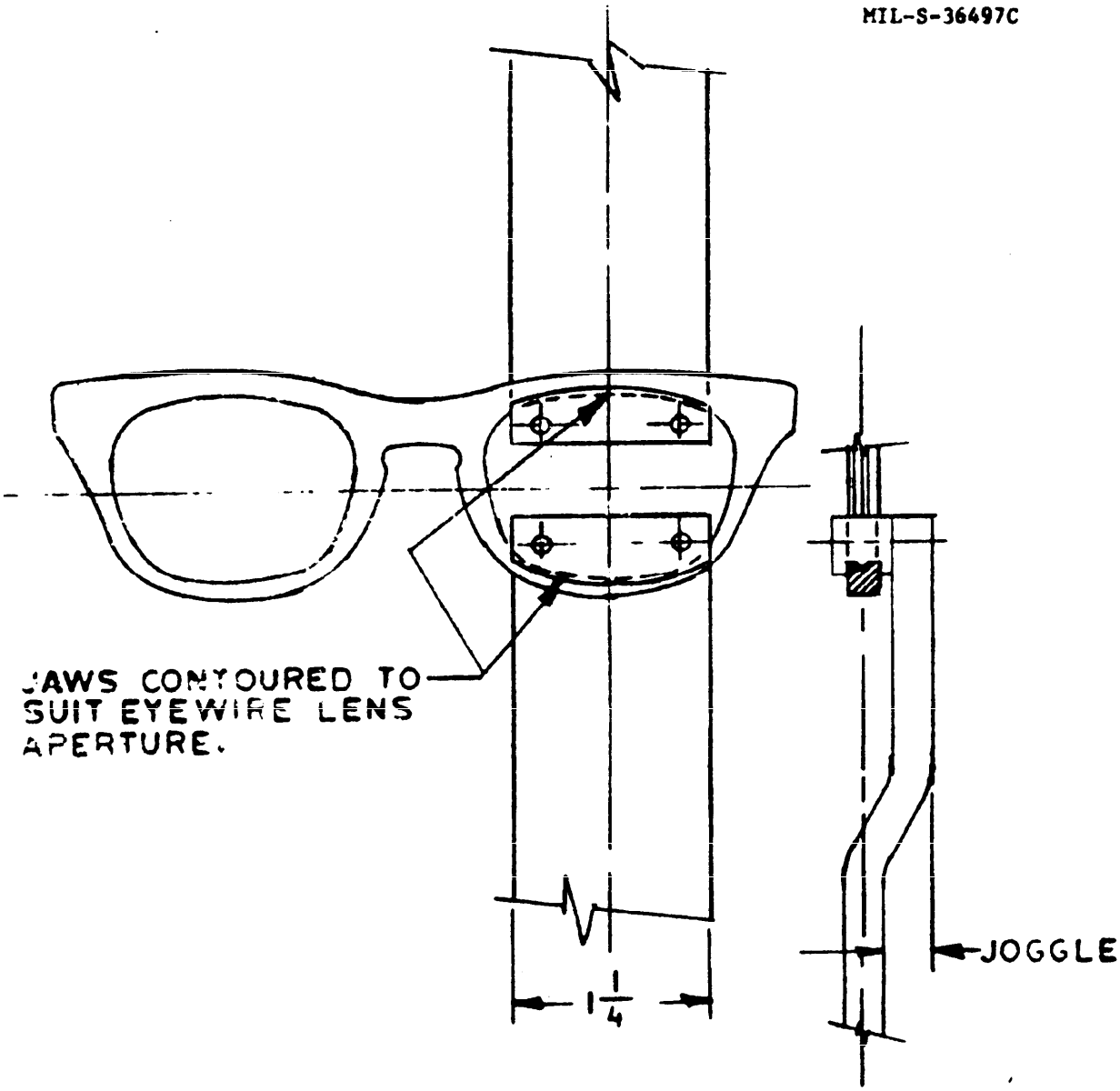
	National Stock Number	Item Identification
Type I	6540-00-482-4292 thru 6540-00-482-4295 and 6540-00-482-9387 thru 6540-00-482-9398	Front, Spectacle, Cellulose Acetate, Black
Type II	6540-00-181-8086 and 6540-00-482-4296 thru 6540-00-482-4299	Temple, Spectacle, Cellulose Acetate, Paddle Type, Black

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project Number 6540-1040
CS

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EYEWIRE TENSILE TEST FIXTURE

FIGURE 1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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